

# Performance Analysis of Sufficient-Statistics OLS

sufficient-regression design note

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## 1 Cost model

Let  $p$  be the number of columns in the design matrix, including any intercept column. Let  $n$  be the number of existing rows,  $m$  the number of appended rows,  $w$  a rolling-window size, and  $T$  the number of rolling windows evaluated.

This note uses dense linear algebra costs:

$$\text{one row contribution } xx^T = \Theta(p^2), \quad \text{one dense solve or factorization} = \Theta(p^3).$$

The  $\Theta(p^2)$  row cost includes the dominant outer product needed for  $G = X^T X$ ; the  $h = X^T y$  contribution costs only  $\Theta(p)$ . Recomputing sufficient statistics from  $r$  dense rows therefore costs

$$C_{stats}(r, p) = \Theta(rp^2).$$

Computing coefficients after the statistics are available costs  $\Theta(p^3)$  with the default standard dense solver. Thus a complete batch OLS fit from raw rows costs

$$C_{fit}(r, p) = \Theta(rp^2 + p^3).$$

The analysis separates statistic maintenance from coefficient solving. That separation is intentional: sufficient statistics remove repeated scans over rows, but a dense coefficient solve is still present whenever coefficients are requested.

## 2 Append case: one million existing rows plus 1000 new rows

Consider an already summarized data set with

$$n = 1,000,000, \quad m = 1,000.$$

The task is to incorporate  $m$  new rows into the existing OLS state.

### 2.1 Final coefficients requested once

If an implementation discards the old summary and recomputes from all raw rows after the append, the cost is

$$C_{raw,final} = \Theta((n + m)p^2 + p^3).$$

If an implementation keeps sufficient statistics, it only forms the contributions of the  $m$  appended rows and then solves:

$$C_{suff,final} = \Theta(mp^2 + p^3).$$

**Theorem 1** (Append speedup for one final solve). *Ignoring the common solve term, sufficient-statistics append gives a statistics-maintenance speedup of*

$$\Theta\left(\frac{(n+m)p^2}{mp^2}\right) = \Theta\left(\frac{n+m}{m}\right).$$

For  $n = 1,000,000$  and  $m = 1,000$ , this ratio is

$$\frac{1,001,000}{1,000} = 1001.$$

*Proof.* Both methods need the same final dense solve after their statistics are available. The raw method forms  $G$ ,  $h$ , and  $q$  from  $n+m$  rows, while the sufficient-statistics method forms contributions from only  $m$  rows because the old  $n$  rows have already been summarized. Dividing the two statistic-formation costs gives the stated ratio.  $\square$

Including the solve term, the end-to-end ratio is

$$\frac{(n+m)p^2 + p^3}{mp^2 + p^3}.$$

When  $p^3$  is small relative to  $mp^2$ , this approaches 1001 for the given numbers. When  $p$  is so large that  $p^3$  dominates both methods, the end-to-end ratio approaches 1 because both methods spend most of their time in the same solve.

## 2.2 Coefficients requested after every appended row

If coefficients are requested after each of the  $m$  appended rows, a raw refit after append  $j$  scans  $n+j$  rows:

$$C_{raw,each} = \sum_{j=1}^m \Theta((n+j)p^2 + p^3) = \Theta(mnp^2 + m^2p^2 + mp^3).$$

The sufficient-statistics approach updates one row and solves at each step:

$$C_{suff,each} = \Theta(mp^2 + mp^3).$$

**Theorem 2** (Append speedup for solve-after-each-append). *For fixed  $p$ , or whenever repeated row scans dominate dense solves, the sufficient-statistics method improves the append stream from  $\Theta(mn)$  row work to  $\Theta(m)$  row work, a  $\Theta(n)$  statistics-maintenance speedup. With  $n = 1,000,000$ , the leading row-scan factor is one million.*

*Proof.* The raw method revisits approximately  $n$  old rows for each append, so its leading row-statistics term is  $mnp^2$ . The incremental method touches only the new row for each append, so its leading row-statistics term is  $mp^2$ . Dividing gives  $n$ .  $\square$

## 3 Rolling window case: window size 1000

Let the rolling window size be

$$w = 1000.$$

For each slide, one row leaves and one row enters.

A raw rolling implementation that recomputes statistics for each window from the  $w$  active rows costs

$$C_{raw,roll}(T, w, p) = \Theta(T(wp^2 + p^3)).$$

A sufficient-statistics rolling implementation subtracts the departing row contribution and adds the arriving row contribution:

$$G^+ = G - x_{old}x_{old}^T + x_{new}x_{new}^T.$$

The matching  $h$  and  $q$  updates cost no more than the two row outer products, so the rolling sufficient-statistics cost is

$$C_{suff,roll}(T, p) = \Theta(T(p^2 + p^3)).$$

**Theorem 3** (Rolling speedup). *Ignoring the common solve term, the rolling sufficient-statistics method reduces statistic maintenance from  $\Theta(Twp^2)$  to  $\Theta(Tp^2)$ . The statistics-maintenance speedup is therefore*

$$\Theta(w).$$

For  $w = 1000$ , this is a 1000-fold asymptotic reduction in row contribution work.

*Proof.* The raw method processes all  $w$  rows in each of the  $T$  windows. The rolling sufficient-statistics method processes exactly two row contributions per slide: one subtraction and one addition. Constants do not affect  $\Theta$  notation, so the ratio is  $\Theta(w)$ .  $\square$

Including dense solves, the end-to-end cost ratio is approximately

$$\frac{wp^2 + p^3}{p^2 + p^3}.$$

For fixed  $p$ , this is  $\Theta(w)$ . If  $p$  is much larger than the window size, the dense solve can dominate and reduce the observable wall-clock speedup even though the row-statistics work still drops by the factor  $w$ .

## 4 Weighted, ridge, and forgetting costs

The sufficient-statistics formulas preserve the same asymptotic update costs for common OLS variants:

| Variant      | Additional work                                                   | Update cost          |
|--------------|-------------------------------------------------------------------|----------------------|
| Weighted OLS | Multiply the row contribution by $w_i$ .                          | $\Theta(p^2)$        |
| Ridge        | Add fixed penalty before solving.                                 | Solve: $\Theta(p^3)$ |
| Forgetting   | Scale $G$ by $\alpha$ , scale $h$ and $q$ , then add the new row. | $\Theta(p^2)$        |

The default dense solver path materializes and adds a  $p \times p$  penalty matrix, so the ridge pre-solve setup is  $\Theta(p^2)$  even when the penalty is diagonal. This term is lower order than the dense  $\Theta(p^3)$  solve, but it is still the implementation's actual setup cost.

Exponential forgetting does not need to rescan historical rows. At time  $t$ , all historical decay is represented by scaling the previous statistics:

$$G_t = \alpha G_{t-1} + x_t x_t^T, \quad h_t = \alpha h_{t-1} + x_t y_t, \quad q_t = \alpha q_{t-1} + y_t^2.$$

Scaling  $G$  is  $\Theta(p^2)$ , matching the row outer-product order.

## 5 Rank-one Cholesky factor prototype

The append-only rank-one prototype targets the solve term directly. Let

$$A = G + R,$$

where  $R$  is the fixed ridge penalty matrix for the estimator. Once the implementation has a Cholesky factor

$$A = LL^T,$$

an appended weighted row  $(x, y, w)$  changes the solve system by

$$A^+ = A + uu^T, \quad u = \sqrt{w}x.$$

A Cholesky rank-one update computes  $L^+$  satisfying

$$A^+ = L^+(L^+)^T$$

in  $\Theta(p^2)$  time. The right-hand side still updates as

$$h^+ = h + wxy,$$

which costs  $\Theta(p)$  once the row is available. Solving

$$Lz = h, \quad L^T\beta = z$$

by triangular substitution costs  $\Theta(p^2)$ .

Therefore, after the first successful factorization, an append followed by coefficients costs

$$\Theta(p^2)$$

for factor maintenance and triangular solves instead of the default  $\Theta(p^3)$  dense solve. This prototype deliberately covers only append updates. Rolling-window downdates would require subtracting  $uu^T$ , which can destroy positive definiteness; supporting that path requires an explicit downdate rejection policy rather than a quiet fallback to a dense solve.

## 6 Memory complexity

A batch fit from raw dense data stores  $X$  and  $y$  with memory

$$\Theta(np + n).$$

An incremental sufficient-statistics fit stores

$$G, h, q, n,$$

with memory

$$\Theta(p^2 + p + 1) = \Theta(p^2).$$

Rolling windows need the summary plus enough information to know the departing row. Storing the active dense rows costs

$$\Theta(wp + w),$$

so rolling memory is

$$\Theta(p^2 + wp + w).$$

This is still independent of the total stream length.

## 7 Interpretation

For the million-plus-1000 append case, the sufficient-statistics advantage comes from not rereading the one million previously summarized rows. For the 1000-window rolling case, the advantage comes from replacing each full-window rebuild with one subtract and one add.

The headline asymptotic speedups are exact for statistic maintenance: 1001 for a final million-plus-1000 append rebuild,  $\Theta(n)$  for solve-after-each append streams, and  $\Theta(1000)$  for a rolling window of size 1000. End-to-end wall-clock speedups in the default estimators are bounded by the unchanged dense solve cost. The append-only Cholesky prototype removes that dense solve from the hot path after the first successful factorization.